

INDEX

NAME Prakhar Patel STANDAL X D SUBJECT Mathematics

Exercise - 11

1. Express each number as a product of its prime factors

(i) 140

$$\begin{array}{r|l}
 2 & 140 \\
 \hline
 2 & 70 \\
 \hline
 5 & 35 \\
 \hline
 7 & 7 \\
 \hline
 1 & 1 \\
 \hline
 \end{array}$$

$$140 = 2 \times 2 \times 5 \times 7$$

$$\underline{\underline{2^2 \times 5 \times 7}}$$

(ii) 156

$$\begin{array}{r|l}
 2 & 156 \\
 \hline
 2 & 78 \\
 \hline
 3 & 39 \\
 \hline
 13 & 13 \\
 \hline
 1 & 1 \\
 \hline
 \end{array}$$

$$156 = 2 \times 2 \times 3 \times 13$$

$$\underline{\underline{2^2 \times 3 \times 13}}$$

(iii) 3825

$$\begin{array}{r|l}
 5 & 3825 \\
 \hline
 5 & 745 \\
 \hline
 13 & 149 \\
 \hline
 13 & 13 \\
 \hline
 1 & 1 \\
 \hline
 \end{array}$$

$$3825 = 3 \times 3 \times 5 \times 5 \times 17$$

$$\underline{\underline{3^2 \times 5^2 \times 17}}$$

(iv) 5005
 $= 5 \times 17 \times 11 \times 13$

(vi) 7429

$$\begin{array}{r|l} 17 & 7429 \\ \hline 19 & 437 \\ \hline 23 & 23 \\ \hline & 1 \end{array}$$

$$7429 = 17 \times 19 \times 23$$

2. Find the LCM and HCF of the following pairs of integers and verify that $\text{LCM} \times \text{HCF} = \text{Product of the two numbers}$

(i) 26 and 91

$$26 = 1 \times 2 \times 13 \times 7^0$$

$$91 = 1 \times 13 \times 7 \times 2^0$$

$$\begin{aligned} \text{HCF} &= 2^0 \times 7^0 \times 13 \\ &= \underline{\underline{13}} \end{aligned}$$

$$\begin{aligned} \text{LCM} &= 2 \times 13 \times 7 \\ 14 \times 13 &= 182 \end{aligned}$$

$$\text{LCM} \times \text{HCF} = 26 \times 91$$

$$13 \times 182 = 26 \times 91$$

$$2366 = 2366$$

(ii) 510 and 92

$$510 = 2 \times 5 \times 17 \times 3 \times 23$$

$$92 = 2^2 \times 23 \times 17 \times 5$$

$$\text{HCF} = 2 \times 5 \times 17 \times 23 \times 3$$

$$\text{LCM} = 2^2 \times 23 \times 17 \times 3 \times 5$$

$$4 \times 23 \times 17 \times 35$$

$$= 23460$$

$$510 \times 92 = 2 \times 23460$$

$$46920 \neq 46830$$

(iii) 336 and 54

$$336 = 2^4 \times 3 \times 7$$

$$54 = 2 \times 3^3 \times 7$$

$$\text{HCF} = 2 \times 3 \times 7$$

$$\underline{6}$$

$$\text{LCM} = 2^4 \times 3^3 \times 7$$

$$16 \times 27 \times 7 = 3024$$

$$3024 \times 6 = 336 \times 54$$

$$18144 = 18144$$

3. Find the LCM and HCF of the following integers by applying the prime factorisation method.

(i) 12, 15 and 21

$$12 = 2 \times 2 \times 3 \times 5^0 \times 7^0$$

$$15 = 5 \times 3 \times 7^0 \times 2^0$$

$$21 = 7 \times 3 \times 5^0 \times 2^0$$

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7$$

$$420$$

$$\text{HCF} = 3 \times 2^0 \times 7^0 \times 5^0$$

$$= \underline{\underline{3}}$$

(ii) 17, 23 and 29

$$17 = 1 \times 17$$

$$23 = 1 \times 23$$

$$29 = 1 \times 29$$

$$\text{HCF} = \underline{\underline{1}}$$

$$\begin{aligned} \text{LCM} &= 17 \times 23 \times 29 \\ &= \underline{\underline{11339}} \end{aligned}$$

(iii) 8, 9 and 25

$$\begin{aligned} 8 &= 2 \times 2 \times 2 \times 5^0 \times 3^0 \\ 9 &= 3 \times 3 \times 5^0 \times 2^0 \\ 25 &= 5 \times 5 \times 3^0 \times 2^0 \end{aligned}$$

$$\text{HCF} = 2^0 \times 3^0 \times 5^0$$

$$\underline{\underline{1}}$$

$$\begin{aligned} \text{LCM} &= 2^3 \times 3^2 \times 5^2 \\ &= 8 \times 9 \times 25 \\ &\Rightarrow \underline{\underline{1800}} \end{aligned}$$

4. Given that $\text{HCF}(306, 657) = 9$, find $\text{LCM}(306, 657)$

$$\text{HCF of } 306 \text{ \& } 657 = 9$$

$$\text{HCF} \times \text{LCM} = P \times Q$$

$$9 \times x = 306 \times 657$$

$$x = \frac{306 \times 657}{9}$$

$$\begin{array}{r} \cancel{10} \cancel{2} \cancel{8} \cancel{3} \\ 34 \end{array}$$

$$\underline{\underline{\text{LCM} = 22338}}$$

5. Check whether 6^n can end with the digit 0 for any natural number n .

$$6^n = (3 \times 2)^n$$

$$6^n = 3^n \times 2^n$$

The numbers which ends with zero have always unique prime factors as 2 & 5, $\rightarrow$

Hence 6 has 3 & 2 as its factors & can never end with zero.

6. Explain why $7 \times 11 \times 13 + 13$ and $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$ are composite numbers

$$7 \times 11 \times 13 + 13$$

$$13 [7 \times 11 \times 1 + 1]$$

$$13 [77 + 1]$$

$$13 \times 78$$

$$13 \times 2 \times 3 \times 13 \times 1$$

Since the given no has more than factors (13, 2, 3) it is a composite number

$$7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5$$

$$5 [7 \times 6 \times 1 \times 4 \times 3 \times 2 \times 1 + 1]$$

$$5 [42 \times 24 + 1]$$

5 [1008 + 1]

7. There is a circular path around a sports field. Sonia takes 18 minutes to drive one round of the field, while Ravi takes 12 minutes for same. Suppose they both start at same point at the same time, and go in same direction. After how many minutes will they meet again at the starting point?

Solution:

Time Sonia took to cover one round = 18 Minutes

// Ravi // = 12 minutes

LCM of 18 & 12

$$18 = 2 \times 3 \times 3$$

$$12 = 2 \times 2 \times 3$$

$$2^2 \times 3^2$$

$$\underline{4 \times 9 = 36}$$

After 36 minutes they will meet again at starting point

Exerc 98e-12

1. Prove that $\sqrt{5}$ is irrational.

Proof: If possible let us assume that $\sqrt{5}$ is not irrational number.

So, it is a rational number

$$\sqrt{5} = \frac{p}{q} \quad \left\{ \begin{array}{l} p \text{ \& } q \text{ are co-primes, } q \neq 0 \\ p, q \in \mathbb{Z} \end{array} \right.$$

Square on both sides

$$(\sqrt{5})^2 = \left(\frac{p}{q}\right)^2$$

$$5 = \frac{p^2}{q^2}$$

$$p^2 = 5q^2 \quad \text{--- (1)}$$

5 is the factor of p^2

5 is also the factor of p

$$p = 5 \times m \quad | \quad p = 5m$$

$$(p)^2 = (5m)^2$$

$$p^2 = 25m^2$$

25 is the factor of p^2

" also of p

eqn (1) and (2)

$$\cancel{5}q^2 = \cancel{25}m^2$$

$$q^2 = 5m^2 \quad \text{--- (3)}$$

5 is the factor of q^2

From above we can say that 5 is the common factor of p & q

which is not possible as p & q being co-primes

This contradiction arises due to our wrong assumption that $\sqrt{5}$ is not irrational.

Hence $\sqrt{5}$ is irrational

$\therefore$ Hp

2. Prove that $3 + 2\sqrt{5}$ is irrational.

Proof: If possible, let us assume that $3 + 2\sqrt{5}$ is not irrational no.

So, $3 + 2\sqrt{5}$ is rational

$$3 + 2\sqrt{5} = \frac{p}{q}$$

Where p, q are co-primes, $q \neq 0$
 $p, q \in \mathbb{Z}$

$$3 + 2\sqrt{5} = \frac{p}{q}$$

$$2\sqrt{5} = \frac{p-3}{q}$$

$$2 \times \sqrt{5} = \frac{p-3}{q}$$

$$\sqrt{5} = \frac{p-3}{2q}$$

$$\sqrt{5} = \frac{p-3}{2q}$$

Hence, in L.H.S. there is an irrational no. and in R.H.S. there are rational number. p, q being integers,

irrational no. can never be equal to a rational no.

This contradiction arises due to our wrong assumption that $3 + 2\sqrt{5}$ is not irrational.

Hence, $3 + 2\sqrt{5}$ is irrational

3. Prove that the following are irrational

(i) $\frac{1}{\sqrt{2}}$

Proof: If possible let us assume that $\frac{1}{\sqrt{2}}$ is not irrational number.

So, $\frac{1}{\sqrt{2}}$ is rational

$$\frac{1}{\sqrt{2}} = \frac{p}{q} \quad \left\{ \begin{array}{l} p, q \in \mathbb{Z}, q \neq 0 \\ p, q \text{ are co-primes} \end{array} \right.$$

$$q = \sqrt{2} p$$

Square on both sides

$$q^2 = 2p^2 \quad \text{--- (1)}$$

2 is the factor of p^2

2 is also the factor of p

$$p = 2 \times m \quad [m \text{ is a some integer}]$$

$$q^2 = 4m^2 \quad \text{--- (2)}$$

4 is the factor of q^2

4 is also the factor of q

from (1) and (2)

$$1 \times 2p^2 = 4m^2$$

$$p^2 = 2m^2$$

$$[p^2 = 2m^2]$$

2 is the factor of p^2
2 is also the factor of p

From above we can say that
2 is the common factor of p & q

which is not possible as p & q
are co-primes.

This contradiction arises due to our
wrong assumption that $\frac{1}{\sqrt{2}}$ is not
irrational.

Hence, $\frac{1}{\sqrt{2}}$ is irrational.

(iii) $\sqrt{5}$

Proof: If possible let us assume that
 $\sqrt{5}$ is not irrational.

So, $\sqrt{5}$ is rational.

$$\sqrt{5} = \frac{p}{q} \quad \begin{cases} q \neq 0, p, q \in \mathbb{Z}, \\ p, q \text{ are co-primes} \end{cases}$$

$$5 \times \sqrt{5} = \frac{p}{q}$$

$$\sqrt{5} = \frac{p}{5q}$$

Here, in L.H.S. there is an irrational no
and in R.H.S. there is rational as p & q
being integers.

Irrational number can never be equal
to rational no.

This contradiction arises due to our
wrong assumption that $\sqrt{5}$ is not
irrational.

Hence, $\sqrt{5}$ is irrational.

(iii) $6 + \sqrt{2}$

Proof: If possible let us assume that
 $6 + \sqrt{2}$ is not irrational.

So, $6 + \sqrt{2}$ is rational.

$$6 + \sqrt{2} = \frac{p}{q} \quad \begin{cases} q \neq 0, p, q \in \mathbb{Z}, \\ p, q \text{ co-primes} \end{cases}$$

$$6 + \sqrt{2} = \frac{p}{q}$$

$$\sqrt{2} = \frac{p-6}{q}$$

Here in L.H.S there is irrational no. and in R.H.S there is rational as p, q being integer.

irrational no. can never be equal to rational.

This contradiction arises due to our wrong assumption that $6 + \sqrt{2}$ is not irrational.

Hence, $6 + \sqrt{2}$ is irrational.